

Mini Project: Disease Risk Prediction System

Objective

Develop a machine learning model to predict disease risk using patient clinical data. Students will implement a **baseline model**, improve it using advanced techniques, and compare the results using standard evaluation metrics.

Dataset

- Heart Disease Dataset
 - URL: <https://www.kaggle.com/datasets/redwankarimsony/heart-disease-data>
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Project Requirements

1. Data Exploration

- Load and inspect the dataset.
- Handle missing values (if any).
- Perform exploratory data analysis (EDA).
- Visualize key patterns using appropriate plots.

2. Data Preprocessing

- Prepare features and target variable.
- Encode categorical variables (if required).
- Apply feature scaling.
- Split the dataset into training and testing sets.

3. Baseline Model

Select **one** of the following as your baseline classifier:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree

Train and evaluate the baseline model using:

- Accuracy
- Precision

- Recall (Sensitivity)
- Specificity
- F1-score
- ROC-AUC

4. Improved Model

Develop an improved model by applying **one or more** of the following:

- Support Vector Machine (SVM)
- Random Forest
- Hyperparameter Tuning (GridSearchCV)

5. Model Comparison

Compare the **baseline model** and the **improved model** using:

- Accuracy
- Precision
- Recall (Sensitivity)
- Specificity
- F1-score
- ROC-AUC
- Confusion Matrix
- Training Time

Present the comparison using tables and graphs, and justify why the improved model performs better (or explain why it does not).

6. Prediction System

Develop a simple prediction system that accepts patient information as input and predicts the disease risk (Low Risk / High Risk).

Deliverables

- Google Colab Notebook (.ipynb)
- Project Report (PDF)
- Presentation (5–7 slides)